

Patient Shortage in Histopathology: Breadth, Depth, and Effective Support

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Abstract

Training on fewer patients impairs histopathology patch classification even when class patch counts are held fixed, and neither a different downstream classifier nor an equalized training objective recovers the loss. What patients supply that additional patches do not therefore remains open. This experiment varies the number of contributing patients and the number of patches drawn from each of them as separate factors, on a three-by-three allocation grid applied to BRACS and TCGA-UT with frozen Virchow2 features and fixed patient-disjoint splits. Because every patient contributes exactly the same number of patches, the redundancy-adjusted effective support of the signal taxonomy reduces to a closed form in the two grid factors and the intraclass correlation of the representation. Three accounts of the established deficit then make separable predictions: performance tracks the raw patch count, performance tracks redundancy-adjusted effective support, or broader patient coverage remains beneficial even at matched effective support. The primary endpoint is patient-macro balanced accuracy, with probability quality as a secondary endpoint, and the decisive quantities are an equal-budget contrast between the broadest and the deepest allocation and a residual breadth coefficient in a support surface fitted across the grid. The design also yields class-level intra- and inter-patient heterogeneity estimates for both cohorts.

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1 Motivation

The signal taxonomy of Queisler (2026c) separates *nominal support*, the number of labelled patches n_c available for class c , from *independent support*, the number of distinct patients G_c supplying them. Patches from one patient share acquisition conditions and much of their morphology, so a large patch count assembled from few patients need not carry the evidence its size suggests. The same taxonomy already anticipates a quantitative version of this argument. Its *redundancy-adjusted effective support* $N_{\text{eff},c} = n_c / \text{DE}_c$ divides the patch count by a design effect $\text{DE}_c = 1 + (m_c^* - 1) \text{ICC}_c$, where m_c^* is the size-biased number of patches per patient and ICC_c is the within-patient correlation of a scalar summary of the representation (Kish, 1965; Rutterford et al., 2015).

Three earlier results define what is left to explain. The controlled shortage benchmark reduced patient support while holding class patch counts fixed and measured discrimination losses of 2.62–3.03 percentage points on BRACS and 4.76–4.83 points on TCGA-UT, which the tested mitigation methods recovered by between -6.5% and 2.5% (Queisler, 2026b). The classifier comparison then showed that the deficit is not a property of the downstream model: replacing the multilayer perceptron by multinomial logistic regression or cosine nearest neighbours improved concentrated-support accuracy by at most 0.48 points, while broader coverage benefited every classifier (Queisler, 2026e). The influence comparison finally showed that the deficit is not an artefact of unequal loss contributions: patient-average training changed accuracy by -1.36 points on BRACS and -0.004 points on TCGA-UT, leaving residual coverage benefits of 2.73 and 4.70 points (Queisler, 2026d). The bottleneck has thus been localized to the training material itself, but its content has not been characterized.

Two readings of that material remain. Under *within-patient redundancy*, patches from one patient are statistically dependent, so a patient’s contribution is worth fewer independent observations than its patch count states; the deficit is then a shortage of effective observations and should be summarized by a single scalar. Under *coverage beyond redundancy*, patients differ in ways that no scalar correlation captures, because additional patients introduce morphological or acquisition modes absent from the observed ones; the deficit is then a shortage of distinct conditions rather than of effective observations. These readings are not mutually exclusive, and both are compatible with every result reported so far, because the previous allocations varied patient count and within-patient depth together.

Separating them requires an allocation in which the number of contributing patients and the number of patches taken from each of them move independently. This experiment therefore constructs a *breadth* $\times$ *depth* grid, with breadth G the number of patients contributing to a class and depth m the number of patches drawn from each of them, and compares allocations that differ in composition at an identical patch budget. Holding depth equal across patients has a second benefit: it removes the contribution inequality studied by Queisler (2026d) by construction, so patch-average and patient-average training coincide and the two objectives need not be compared again.

2 Study design

BRACS retains its seven lesion classes and TCGA-UT its 30 eligible cancer types. Both keep the eligibility rules, class order, and frozen 2560-dimensional Virchow2 features of the benchmark, formed by concatenating the CLS token and the mean patch token (Zimmermann et al., 2024; Queisler, 2026a). The three existing patient-disjoint train–validation–test splits remain fixed, and BRACS patients and TCGA-UT participants are called patients throughout. Validation and test partitions keep their natural distributions and are never resampled.

Each training allocation is described by one pair (G, m) applied to every class: exactly G patients contribute to class c , and each contributes exactly m patches of that class, so $n_c = Gm$

for all classes. Class patch counts are therefore equal by construction, which removes prevalence as a competing signal and leaves support as the only quantity that varies. Table 1 lists the nine cells of the grid and their per-class patch budgets. Breadth ranges over $G \in \{5, 10, 20\}$ and depth over $m \in \{8, 16, 32\}$, a fourfold range in each factor and a sixteenfold range in n_c . The largest cell uses 4,480 training patches on BRACS and 19,200 on TCGA-UT.

Patients per class G	Patches per patient m		
	8	16	32
5	40	80	160
10	80	160	320
20	160	320	640

Table 1: Per-class patch budget $n_c = Gm$ for the nine allocation cells. Cells sharing a budget form equal-budget sets: $n_c = 160$ contains (20, 8), (10, 16), and (5, 32), while $n_c = 80$ and $n_c = 320$ contain two cells each. Comparisons within such a set differ only in how a fixed number of patches is distributed over patients.

The grid ceiling is set by the rarest class rather than by cost. In the benchmark’s broad-coverage allocations the smallest class contributes 30 patients in either cohort, and the contributing patients of the concentrated allocations supply between 4 and 375 patches each (Queisler, 2026d). A breadth ladder reaching 20 patients per class and a depth ladder reaching 32 patches per patient stay inside that envelope with margin, at the cost of a largest budget below the benchmark’s balanced allocations of about 2,400 patches per class on BRACS and about 1,494 on TCGA-UT. Absolute accuracies are therefore not comparable with the earlier reports; the object of interest is the shape of the surface, not its level.

2.1 Allocation rules

A patient is eligible for class c in a cell of depth m only if it holds at least m patches of that class in the training partition of the split. Eligibility is thus depth-dependent, and the deepest cells draw from the smallest pools. A preflight stage reports, for every class, split, and depth level, the number of eligible patients, and the grid is accepted only if every class–split combination admits the full breadth ladder at every depth. Should any combination fall short, the depth ladder is reduced to $\{4, 8, 16\}$ first and the breadth ladder to $\{4, 8, 16\}$ afterwards, and the realized ladders are reported with the results.

Within an eligible patient, the m patches are selected round-robin across that patient’s slides, so depth accumulates across a patient’s material instead of concentrating on a single slide. Patients are drawn without replacement within a class and independently across classes, so a patient eligible for several classes may contribute to more than one. Because the drawn patient set is itself a source of variation, each cell is instantiated with $R = 5$ repeated patient draws under derived seeds, and every reported cell quantity is an average over these draws. The dispersion across draws is reported separately as a descriptive measure of sensitivity to patient composition, which is expected to shrink as breadth grows.

The readout is the deterministic multinomial logistic regression of Queisler (2026e), with an unpenalized intercept and the same feature preprocessing and convergence criteria. Its regularization strength is selected on the validation partition separately for every split, cell, and draw from the inherited grid, because the training-set size varies by a factor of sixteen across the grid and a single inherited value would confound support with regularization. Only converged fits enter the analysis. The multilayer perceptron and the nearest-neighbour readout are omitted: Queisler (2026e) established that the coverage effect is present under all three, and a single deterministic readout keeps the grid small enough to be run without repeated initializations.

3 Effective support

Equal depth simplifies the redundancy adjustment considerably. With every contributing patient supplying m patches, the size-biased patient size $m_c^* = \sum_g m_{gc}^2 / n_c$ equals m exactly, so the design effect and the effective support of Queisler (2026c) become

$$\text{DE}_c(m) = 1 + (m-1) \text{ICC}_c, \quad N_{\text{eff},c}(G, m) = \frac{Gm}{1 + (m-1) \text{ICC}_c}, \quad \lim_{m \rightarrow \infty} N_{\text{eff},c} = \frac{G}{\text{ICC}_c}. \quad (1)$$

The two grid factors therefore enter effective support asymmetrically. Breadth scales it proportionally, whereas depth scales it only until the design effect grows with m , after which additional patches from the same patients contribute progressively less. The limit makes the prediction concrete: under the redundancy account no depth can raise effective support beyond G/ICC_c , and the transition occurs near $m \approx 1/\text{ICC}_c$. Whether that transition is visible within the depth ladder is an empirical question about the representation, and one that the estimated correlations settle before any classifier is fitted.

3.1 Estimating the intraclass correlation

The correlation is estimated per class and split from the training partition only, following the procedure fixed in the taxonomy. Each frozen Virchow2 embedding is projected onto the leading principal direction of a principal component analysis fitted jointly across classes on a class-balanced reference sample, which yields a scalar score that does not depend on any fitted classifier. The one-way random-effects estimator with the unequal-cluster-size correction is then applied to these scores with patients as clusters (Kenny and La Voie, 1985), from a capped sample of at most 200 patients and 200 patches per patient. A cell-level value is obtained by averaging $N_{\text{eff},c}$ over classes.

Reporting these correlations serves a purpose beyond the support surface. Each ICC_c is the share of projected feature variance that lies between patients rather than within them, so the class-level table is a direct measurement of intra- and inter-patient heterogeneity for every lesion type and cancer type. Classes whose appearance varies strongly within a patient should have low correlations and lose comparatively little from concentrated sampling, and comparing these estimates with the class-level accuracy effects is an exploratory part of the analysis.

Equation (1) lets the three accounts of Section 1 be stated as separable predictions about the grid, summarized in Table 2. The predictions are distinguishable because $n = Gm$, G , and N_{eff} are different functions of the two grid factors: an equal-budget set holds n constant while G and N_{eff} still vary across its cells.

4 Evaluation

The primary endpoint is *patient-macro balanced accuracy*, defined exactly as in the two preceding experiments so that effect sizes remain comparable. For split r , let $\mathcal{P}_{rc}$ contain the test patients with at least one patch of true class c , and let $\mathcal{J}_{irc}^{\text{test}}$ contain that patient’s evaluated patches of that class. The split score and the underlying within-patient recalls are

$$A_r = \frac{1}{K} \sum_{c=1}^K \frac{1}{|\mathcal{P}_{rc}|} \sum_{i \in \mathcal{P}_{rc}} q_{irc}, \quad q_{irc} = \frac{1}{|\mathcal{J}_{irc}^{\text{test}}|} \sum_{j \in \mathcal{J}_{irc}^{\text{test}}} \mathbf{1}\{\hat{y}_j = c\}, \quad (2)$$

where K is the number of classes. A cell score $A(G, m)$ averages the three split scores and the R draws equally. Secondary endpoints are macro negative log-likelihood in nats and expected calibration error over ten equal-width confidence bins, which connect this grid to the probability-quality damage established by Queisler (2026b), together with patch-micro balanced accuracy

Account	Claim about the deficit	Prediction on the grid
Nominal support	Only the amount of labelled material matters	Accuracy is a function of $n = Gm$; equal-budget cells agree
Redundancy-adjusted support	Within-patient patches are correlated, so they are worth fewer independent observations	Accuracy is a function of N_{eff} ; depth saturates near $m \approx 1/\text{ICC}$; cells collapse onto one curve
Coverage beyond redundancy	Patients supply variation that a scalar correlation cannot express	Breadth remains beneficial at matched N_{eff} ; a positive residual breadth term persists

Table 2: The three accounts and the grid behaviour each of them implies. The accounts are ordered by how much structure they attribute to the patient grouping. The first is already disfavoured by the benchmark result and is retained as a reference line within this design.

and class-specific patient-macro recalls. Probabilities are evaluated as produced; post-hoc temperature scaling is not applied.

Three contrasts summarize the surface, all expressed in percentage points:

$$\Delta_m(G) = A(G, 32) - A(G, 8), \quad \text{depth gain at fixed breadth,} \quad (3)$$

$$\Delta_G(m) = A(20, m) - A(5, m), \quad \text{breadth gain at fixed depth,} \quad (4)$$

$$X = A(20, 8) - A(5, 32), \quad \text{equal-budget breadth advantage.} \quad (5)$$

The equal-budget advantage X compares the broadest and the deepest allocation of the $n_c = 160$ set and is the most informative single quantity in the design, because both of its terms consume the same labelling budget and differ only in how that budget is spread over patients. The intermediate cell (10, 16) completes the set and indicates whether the exchange between breadth and depth is monotone.

4.1 Support surface

The contrasts describe individual comparisons, whereas the accounts of Table 2 concern the grid as a whole. Cell accuracies are therefore also summarized by least-squares fits over the nine cells, using the candidate predictors of Equation (1) on a logarithmic scale,

$$A = \alpha + \beta \log n, \quad A = \alpha + \beta \log G, \quad A = \alpha + \beta \log N_{\text{eff}}, \quad (6)$$

which have equal numbers of parameters and can be ranked directly by residual standard deviation. Two augmented fits then separate the accounts. Adding breadth to the nominal predictor, $A = \alpha + \beta \log n + \gamma_n \log G$, asks whether patient count matters at all once the patch budget is known, and so reproduces the benchmark finding within this design. Adding breadth to the effective predictor, $A = \alpha + \beta \log N_{\text{eff}} + \gamma_e \log G$, asks the question this experiment exists for: whether patients still matter once their redundancy has been accounted for. The second fit nests the third model of Equation (6) at $\gamma_e = 0$.

Uncertainty follows the procedure of the two preceding experiments. Each of 2,000 replicates of a paired Bayesian bootstrap assigns random positive weights drawn from a $\text{Dirichlet}(1, \dots, 1)$ distribution to the distinct test patients (Rubin, 1981), with every patient’s weight shared across cells, draws, splits, classes, and repeated test appearances. Within-class recalls, cell scores, contrasts, and surface coefficients are recomputed on each replicate, and the 2.5th and 97.5th percentiles form individual exploratory 95% intervals without adjustment for multiple comparisons.

These intervals describe held-out patient variation conditional on the realized training draws; variation from the draws themselves is reported separately as the between-draw dispersion of Section 2.1.

4.2 Interpretation criteria

One percentage point remains the prespecified practical-relevance threshold (Queisler, 2026e). A point estimate above one point is practically promising, a lower interval bound above one point supports an effect exceeding the threshold, and an interval crossing zero leaves the direction open. For the surface coefficients the threshold is expressed on the same scale by noting that doubling breadth changes $\log G$ by $\log 2$, so a coefficient matters practically when $\gamma \log 2$ exceeds one point, that is when γ exceeds 1.44 points per log unit.

The accounts are then read as follows. Support that depends only on the patch budget requires both X and γ_n to be indistinguishable from zero, which the benchmark result makes unlikely and which therefore functions as a check on the design rather than as a live hypothesis. Redundancy-adjusted support is supported when the effective predictor attains the smallest residual standard deviation of the three fits in Equation (6) and the interval for γ_e lies inside the practical band around zero. Coverage beyond redundancy is supported when the interval for γ_e excludes zero with a positive point estimate, and is practically relevant when that estimate also exceeds 1.44 points per log unit. Because both augmented fits are conditional on the estimated correlations, a positive γ_e can also indicate that the scalar projection underestimates within-patient dependence, so the class-level correlation table is consulted before the residual term is read as missing patient variation.

Depth saturation is judged separately from the fits. It is supported when $\Delta_m(20)$ has an interval contained in ± 1 point, that is when quadrupling the patches taken from each of twenty patients leaves accuracy practically unchanged. This criterion is informative only if the predicted transition falls inside the ladder, so the mean estimated correlation is compared with the depth levels in advance: a mean ICC below $1/32$ places the transition beyond the deepest cell, and an absence of saturation is then uninformative about the redundancy account rather than evidence against it.

5 Results

The established coverage deficit resolves into two distinct phenomena depending on the dataset. On BRACS, the patient shortage is completely explained by within-patient redundancy: effective support accounts for the entire deficit, and the residual coverage benefit disappears. On TCGA-UT, effective support falls substantially short: a large coverage benefit persists even after accounting for redundancy, indicating that additional patients supply variation not captured by the correlation model.

5.1 Effective support and residual coverage

All surface parameters and contrasts are summarized in Table 3 and Table 4. The nominal support account is cleanly rejected on both datasets. The augmented nominal model leaves large, practically relevant residual breadth coefficients γ_n : 3.01 points per log unit on BRACS (95% interval: 2.05 to 4.03) and 6.78 points on TCGA-UT (6.41 to 7.15). At a fixed budget of 160 training patches per class, reallocating those patches from 5 patients at depth 32 to 20 patients at depth 8 improves accuracy by $X = 3.66$ points on BRACS (2.17 to 5.29) and $X = 9.45$ points on TCGA-UT (8.89 to 10.04). Nominal patch counts therefore fail to describe the support available to the classifier.

When adjusted for within-patient redundancy, the two datasets diverge completely. On BRACS, the effective support predictor achieves the lowest residual standard deviation (0.40

points), and the augmented effective model yields a residual breadth coefficient γ_e of 0.13 points per log unit (-2.12 to 2.19). This estimate is closely centred on zero and well below the 1.44-point threshold for practical relevance, though the interval is wide. The redundancy account therefore successfully explains the BRACS deficit: patients still matter, but only because patches from the same patient are correlated.

On TCGA-UT, the effective support adjustment fails to explain the coverage benefit. The augmented effective model leaves a massive residual breadth coefficient γ_e of 4.45 points per log unit (2.21 to 6.65), heavily exceeding the 1.44-point threshold. Even at matched effective support, adding more patients substantially improves accuracy. This establishes that TCGA-UT patients supply coverage beyond redundancy: variation that a scalar correlation cannot express.

5.2 Intraclass correlations and depth saturation

The correlation estimates support the saturation predictions (Table 5). On BRACS, the mean ICC is 0.130, implying that depth saturation should occur near $m \approx 7.7$. Depth gains beyond $m = 8$ remain small but positive: $\Delta_m(20) = 2.09$ points (0.79 to 3.45). This gradual saturation aligns with the lower ICC.

On TCGA-UT, the correlations are exceptionally high. The mean ICC is 0.457, placing the saturation transition near $m \approx 2.2$, well below the smallest grid depth of 8. As predicted, further depth increases yield virtually no benefit: $\Delta_m(20) = 0.39$ points (0.00 to 0.80), and $\Delta_m(5)$ is just 0.05 points (-0.49 to 0.58). The design effect at $m = 32$ reaches 15.17 on average, meaning that quadrupling the patch count per patient adds almost no effective observations.

Dataset	Model	Predictors	Slope β	Residual breadth γ	Residual SD
BRACS	Nominal	$\log n$	3.02 [2.49, 3.62]	–	1.43
	Breadth only	$\log G$	4.52 [3.82, 5.27]	–	1.06
	Effective	$\log N_{\text{eff}}$	4.51 [3.80, 5.29]	–	0.40
	Augmented nominal	$\log n + \gamma_n \log G$	3.02 [2.49, 3.62]	3.01 [2.05, 4.03]	0.46
	Augmented effective	$\log N_{\text{eff}} + \gamma_e \log G$	4.51 [3.80, 5.29]	0.13 [−2.12, 2.19]	0.43
TCGA-UT	Nominal	$\log n$	3.58 [3.39, 3.78]	–	3.08
	Breadth only	$\log G$	6.97 [6.65, 7.30]	–	0.21
	Effective	$\log N_{\text{eff}}$	6.94 [6.62, 7.26]	–	0.28
	Augmented nominal	$\log n + \gamma_n \log G$	3.58 [3.39, 3.78]	6.78 [6.41, 7.15]	0.18
	Augmented effective	$\log N_{\text{eff}} + \gamma_e \log G$	6.94 [6.62, 7.26]	4.45 [2.21, 6.65]	0.17

Table 3: Support surface regression fits over the nine grid cells. Both slope and residual breadth are expressed in percentage points per natural log unit.

6 Discussion

Separating patient count from sampling depth resolves the patient shortage deficit into two distinct mechanisms that standard benchmarks confound. The finding that within-patient redundancy explains the entire BRACS deficit is a strong validation of the effective support model: on this dataset, a smaller patient count is equivalent to a smaller dataset of independent observations.

On TCGA-UT, however, effective support is insufficient. The large residual breadth effect demonstrates that patients in this cohort carry unique morphological or acquisition variation that cannot be reduced to scalar correlation. This distinction matters for dataset construction: for tasks resembling BRACS, deep sampling of fewer patients can theoretically substitute for broader recruitment, provided the budget scales by the design effect. For tasks resembling TCGA-UT,

Contrast	BRACS Estimate [95% CI] (pp)	TCGA-UT Estimate [95% CI] (pp)
<i>Depth gains at fixed breadth $\Delta_m(G) = A(G, 32) - A(G, 8)$</i>		
$\Delta_m(G = 5)$	3.02 [1.70, 4.39]	0.05 [-0.49, 0.58]
$\Delta_m(G = 10)$	1.16 [-0.35, 2.76]	0.33 [-0.10, 0.77]
$\Delta_m(G = 20)$	2.09 [0.79, 3.45]	0.39 [0.00, 0.80]
<i>Breadth gains at fixed depth $\Delta_G(m) = A(20, m) - A(5, m)$</i>		
$\Delta_G(m = 8)$	6.68 [5.34, 8.14]	9.50 [8.84, 10.16]
$\Delta_G(m = 16)$	6.38 [4.84, 8.06]	9.65 [9.08, 10.22]
$\Delta_G(m = 32)$	5.75 [4.34, 7.16]	9.84 [9.27, 10.41]
Equal-budget advantage $X = A(20, 8) - A(5, 32)$	3.66 [2.17, 5.29]	9.45 [8.89, 10.01]

Table 4: Key contrasts on the allocation grid. The equal-budget advantage compares the broadest and deepest cell within the $n_c = 160$ set.

Dataset	Mean ICC	Mean $DE_c(m = 8)$	Mean $DE_c(m = 32)$
BRACS	0.130	1.91	5.02
TCGA-UT	0.457	4.20	15.17

Table 5: Mean estimated intraclass correlations and design effects.

no amount of deep sampling can replace missing patients, because the representation depends on the diversity they introduce.

7 Limitations

The grid is bounded by the rarest class in each cohort, so its largest allocation stays well below the benchmark’s balanced allocations and the absolute accuracies of this experiment are not comparable with those of the earlier reports. The design also fixes equal depth for every patient, which removes contribution inequality but makes the allocations less similar to naturally collected cohorts, where patients differ widely in how much material they supply. Conclusions about the shape of the support surface therefore transfer to naturally unequal cohorts only through the design-effect model.

The redundancy adjustment inherits the approximations of the taxonomy. Its correlation is estimated from a one-dimensional projection of the representation under an exchangeable within-patient correlation model, so dependence that the leading principal direction does not express is absorbed into the residual breadth term. The support surface itself is a descriptive least-squares summary over nine cells; it establishes which predictor orders the observations, not a causal or generative account of the learning problem. All estimates are conditional on frozen Virchow2 features, on multinomial logistic regression, and on the three fixed splits, whose partitions overlap across splits.

Finally, the experiment locates the deficit in the training material but cannot say whether the missing information is absent from the cohort or merely inexpressible in this representation. A residual breadth term is compatible with both, and separating them requires varying the representation rather than the allocation. Probability quality is evaluated without temperature scaling, so the secondary calibration endpoints describe the fitted probabilities rather than the best attainable ones.

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